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CIVA-Net: Cross Modal Attention Fusion for AI Generated Content Detection

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***Abstract*—The emergence of AI-generated information poses a big threat to the validity of information. Synthetic media has become more realistic than ever with the development of the large language models and diffusion-inspired generators, such as GPT-4 and DALL·E. The social media ecosystems also play a significant role in this dilemma because false information becomes viral within networks. Current detection methods tend to concentrate on either unimodal structure or ensemble systems and do not encode the semantic connections between text/visual information. Our research paper is a proposal of a new framework, CIVA-Net (Caption-Image Visual Attention Network), which uses cross-modal attention to learn joint representations on caption-image pairs in order to classify AI-generated content (AIGC). We are using RoBERTa and the text encoder and Vision Transformer (ViT-B/16) as the image encoder. Our central gift is that of a bidirectional cross-attention mechanism that facilitates the information flow between modalities: image features attend to textual tokens, and the text representations attend to visual patches. The architecture is able to capture semantic-visual anomalies that describe misinformation generated by AI. We tested CIVA-Net on the dataset of MiRAGeNews, which consists of 15,000 pairs of captions and images of various generators and news outlets. Our model has 89% accuracy and 96% recision, which is a major improvement over the unimodal baselines as well as the competing multimodal models. The representations learned are clearly separated in the feature space, which confirms the usefulness of cross-modal fusion. Even though domain generalization issues, the detection of complex synthetic material, and related concerns have still not been completely addressed, our findings demonstrate that the hybrid textual-visual modeling of attention-based fusion is an effective tool for AI-generated content detection.**

***Index Terms*—Cross-Modal Fusion, Fake News Detection, Multimodal Learning, AI-Generated Content.**

I. Introduction

The generative AI (GenAI) systems, including GPT-4 and Midjourney, can now create extremely realistic synthetic images, and they are a genuine threat to the virtual information space. Consequently, the synthesized or AI-generated content is now a major challenge to detect, since the common unimodal detectors are not always capable of identifying any latent semantic anomaly between the text and image data. Current AI-generated content (AIGC) detection methods can be divided into three main classes. The first is unimodal methods, which analyze text or images independently using features such as linguistic patterns, n-gram distributions, or visual artifacts. While these yield good results in their respective fields, they fail to capture important information arising from text-image relationships. The second class is multi-stream architectures, which process text and images separately before combining predictions through late fusion or voting processes [20]. Although they use both types of information, they cannot model the fine-grained semantic correspondence between captions and visual content. The third class is the primary fusion methods, which combine different features before classification, but often fail to capture fine-grained cross-modal interactions [23].

Unlike conventional unimodal detectors that rely on pixel-level artifacts, CIVA-Net addresses a new challenge: semantic and visual anomaly detection. We propose a two-way fusion architecture that detects high-quality synthetic images when they do not logically match their captions for example, if a 'crowd' descriptive caption is associated with a scattered or incomplete image. The model keeps the low-level encoders stable and uses an attention-based pooling technique, which allows it to achieve 89% accuracy on the MiRAGeNews dataset, significantly reducing false alarms compared to standard fusion methods.

The first of three key contributions of our research is to propose a new architecture that combines RoBERTa and Vision Transformer encoders through a cross-modal attention layer, and creates a unified representation space for multimodal fake news detection. Second, we conducted comprehensive experiments on the MiRAGeNews benchmark [12], demonstrating that our method achieves 89% accuracy and 96% precision, outperforming unimodal baselines with lower pre-fusion accuracy and competitive multimodal approaches. Third, we provide detailed analysis of the learned representations through visualization techniques, showing clear separation between real and synthetic samples in the feature

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space. These contributions establish cross-modal attention as an effective mechanism for detecting AI-generated content and provide a foundation for future research in multimodal misinformation detection.

II. Related Work

Recent research in misinformation detection has shifted from unimodal approaches to multimodal systems, which combine both language and vision. Vision-language pretraining (VLP) produces robust cross-modal representations for the first time. Inverse models such as CLIP align captions and images in the embedding space for zero-shot transfer learning [4]. Later, refined alignment and fusion were designed: ALBEF implements "fuse before alignment" [5] with momentum distillation, and ViLT removes convolution for faster tokenization [6], and BLIP/BLIP-2 combines the understanding and generation or link frozen encoders through a query transformer [7]. Larger frameworks such as SimVLM, Flamingo, and LLaVA also show that scaling and instruction tuning improve logic [15].

Multimodal detection, especially for fake news and AIGC, leverages this foundation by identifying inconsistencies between text and images. Co-attention networks have become a popular method for modeling the bidirectional interaction between textual and visual content. For example, MCAN stacks co-attention layers to iteratively fuse features from both the image and the caption, enhancing their alignment [8]. Knowledge-aware models send signals in both directions entities and relationships which makes them more effective at detecting anomalies [9]. Already, methods such as progressive fusion and matching-aware co-attention support multi-resolution interactions and distillation, thereby reducing the impact of small perturbations [13], [14]. Some transformer-based models using the ViT backbone have created stronger connections between text and visual features and enabled effective multimodal detection [21], [24]. These techniques indicate that adding bidirectional co-attention mechanisms while maintaining unidirectional strength allows for more reliable fusion of information from different modalities.

Several main techniques are almost always used in multimodal image and text pair detection. The first technique is the use of two-way co-attention and cross-attention, where captions are associated with visual tokens and vice versa, visual information is matched with captions [8], [11]. The second technique is progressive or staged fusion, which makes logical connections from token-level to scene-level, allowing models to better capture the overall context [10], [24]. Third, alignment-aware training, such as using contrast loss, auxiliary heads, or distillation, stabilizes the fused representations and makes the model more robust against distributional changes [5].

These techniques are typically implemented in conjunction with the RoBERTa–ViT pipeline, where unimodal encoders are lightly-tuned and co-attentional mechanisms handle reasoning processes. Nevertheless, strength is a significant issue. Recent efforts have also tried to solve this problem by integrating contrastive learning and optimal transport methods, which enhance the generalization abilities of a wider range of data sets, including Weibo and PHemE [18]. The context of the scene is well realized in the SceneFND model, and this helps the model to identify realistic yet deceiving visual aspects [19].

This paper suggests an original two-way RoBERTa-ViT co-attention to overcome those shortcomings. In this architecture, the caption tokens are concerned with image patches whereas visual tokens are concerned with image captions so that semantic consistency is achieved prior to classification. Reverse alignment and attentive optimization are additional factors that stabilize the learning process. Our approach builds on the advantages of unimodal encoders and offers remedies to edge cases, in which bidirectional attention is introduced as a key process that allows the successful cross-modal validation.

III. Methodology

A. Dataset

We have used the MiRAGeNews dataset [12] as a criterion for multimodal detection. It has 15,000 caption-image pairs with the equivalent class (label 1 = AI-generated, 0 = human-written). Specifically, the official split provides 10,000 training pairs and 2,500 validation pairs, which are collected from paintings created for New York Times articles and Midjourney. In addition, five isolated test subsets (500 pairs each) support both in-domain and out-of-domain evaluation: Test 1 (NYT + Midjourney), Test 2 (BBC + DALL·E), Test 3 (CNN + DALL·E), Test 4 (BBC + Stable Diffusion XL), and Test 5 (CNN + Stable Diffusion XL). As a result, the benchmark is able to transfer both sources to the generator. To determine the incremental value, we present only the text-based baseline before introducing the cross-modal method.

B. Unimodal Baselines

To establish performance benchmarks, we fine-tuned RoBERTa-base and ViT-B/16 independently. Images were resized to 224×224 and normalized, while captions were truncated to 128 tokens. A linear classification head was applied to the final embeddings of both models.

C. Cross-modal model

The overall architecture is detailed in Figure 1. We connected the RoBERTa-Base text encoder to the ViT-B/16 vision encoder using a bidirectional cross-modal attention mechanism. We froze the bottom eight layers of both backbones to retain the generic features learned during pretraining such as low-level textures in ViT and basic syntax in RoBERTa. Training these layers on a dataset of only 15,000 samples would likely cause overfitting. By fine-tuning only the top four layers, we allow the model to adapt to the high-level semantic inconsistencies specific to AI-generated content without destroying the pre-trained weights. Furthermore, we employed average pooling for the text representation rather than the standard [CLS] token. In news captions, key forensic details (like names or locations) can appear anywhere in the sequence. Averaging the valid tokens ensures the cross-modal attention mechanism

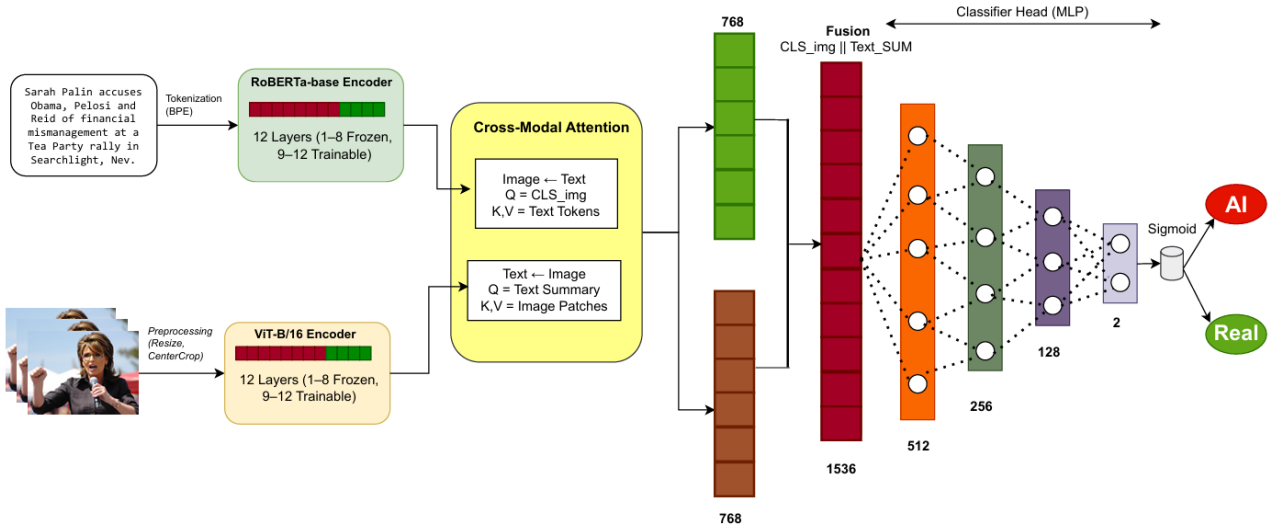

Fig. 1. Architecture of the proposed multimodal detection model. The framework leverages RoBERTa for text encoding and ViT-B/16 for image encoding, followed by cross-modal attention to fuse representations. The fused features are passed through a multi-layer perceptron classifier to distinguish between AI-generated and real content.

considers the full context of the sentence, preventing the model from over-relying on a single classification token which may be unstable in short captions.

Feature Pooling Strategy: Unlike standard BERT classification which relies on the [CLS] token, we employ average pooling over all valid text tokens (B×1×768). We chose this because forensic cues in news captions (specific names, actions) are distributed throughout the sequence. The [CLS] token is often biased towards the start of the sentence, whereas average pooling aggregates the full semantic context, providing a more robust query for the visual attention module.

Bidirectional Attention: These text features serve as queries to attend to visual patches (B×P×768), while image features simultaneously attend to text tokens. This bidirectional flow allows the model to pinpoint exactly where the text contradicts the image. Stable updates are ensured with the help of residual connections, layer normalization, and dropout. As a result, the updated summary is mapped to $B \times 1536$ by a compact MLP (normalization, non-linearity, dropout = 0.3). This bidirectional interaction is conducted through a cross-modal attention process, which enables mutual feature exchange between visual and textual representations.

1) Cross-Modal Attention: Our fusion mechanism is based on the scaled dot-product attention that enables features of images to inform text features and the other way around. Given a query matrix Q, a key matrix K, and a value matrix V, the attention output is calculated as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

In our framework, we have two distinct attention flows:

- **Image-to-Text Attention**: The CLS token in the image is a query, and the text tokens are the instructions to the model on their position and meaning so that the model will choose the most appropriate words or captions on the content of the image.

$$\text{Text to Image} = \text{Attention}(\text{Image}_{\text{CLS}}, \text{Text}_{\text{seq}}, \text{Text}_{\text{seq}}) \tag{2}$$

- **Text-to-Image Attention**: The model is given the average-pooled text embedding as a query, and the image patch tokens are given instructions by the model in the order of their values. This allows the model to identify which visual regions are most relevant to the caption.

$$\text{Image-to-Text Attention} = \text{Attention}(\text{Text}_{\text{summary}}, \text{Image}_{\text{patches}}, \text{Image}_{\text{patches}}) \tag{3}$$

2) Loss Function and Regularization: Throughout the models, optimization has been handled using BCEWithLogitsLoss. This loss combines a sigmoid activation and binary cross-entropy into a single, numerically stable function. For a given predicted logit x and a target label y, the loss is as follows, where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function.

$$L(x, y) = -\left[y \cdot \log(\sigma(x)) + (1 - y) \cdot \log(1 - \sigma(x))\right] \tag{4}$$

Additionally, we employed AdamW to drive parameter updates. AdamW decouples the weight decay from the gradient update, which improves regularization and prevents overfitting. For a parameter θ, the update rule is as follows, where η_t is the learning rate, λ is the weight decay factor, and $\hat{m}_t$ is the bias-corrected first moment estimate (momentum).

$$\theta_{t+1} = \theta_t - \eta_t(\hat{m}_t + \lambda\theta_t) \tag{5}$$

Finally, we utilized Label Smoothing to reduce model overconfidence. Given a ground truth label $y \in \{0, 1\}$ and a smoothing factor ϵ, the smoothed target y_{smooth} is calculated as:

$$y_{\text{smooth}} = y(1 - \epsilon) + 0.5\epsilon \tag{6}$$

This smoothed target is then used in place of the original hard label y in the loss function.

IV. Result Analysis

We present the test results for the text-only, image-only, and multimodal systems. Their performance is summarized in Table I and compared against the baseline models. Accuracy, precision, recall, and F1-score are used as the main evaluation measures. When available, ROC-AUC and PR-AUC are also reported to show the models ability to separate classes. For the main comparison, we focus on accuracy and F1-score.

A. RoBERTa (text modality)

RoBERTa reached an accuracy of 0.8468, with a precision of 0.7777, a recall of 0.9712, and an F1-score of 0.8637. The very high recall shows that the model is highly sensitive to the minority positive class, which helps reduce false negatives. This result is consistent with the ability of pretrained contextual embeddings to capture lexical and syntactic patterns in synthetic captions. However, the lower precision suggests that false positives in the majority class remain significant. This often occurs when human captions use templated or formal phrasing that overlaps with the model's decision boundaries. Given the mild class imbalance, such a gap between precision and recall is expected. Small adjustments to the decision threshold can reduce false positives, increase accuracy, and keep the F1-score at the same level..

B. Vision Transformer (image modality)

The Vision Transformer achieved an accuracy of 0.7336, with precision 0.6805, recall 0.8808, F1-score 0.7678, ROC-AUC 0.8288, and PR-AUC 0.8123. Although validation performance during fine-tuning was strong, the results on the held-out test set were weaker, suggesting a domain shift between training and evaluation images. As expected for pixel-based models, performance is affected by generator diversity, post-processing, and scene composition. High-quality renderers may reduce high-frequency cues, while realistic textures and lighting can blur class boundaries, leading to more false negatives. Compared to the text-only model, recall remains competitive, but precision is lower, showing that without linguistic cues the model struggles to clearly separate photorealistic synthetic images from real ones.

C. Multimodal Fusion (text + image)

The multimodal fusion model, which seamlessly combines text and visual representations, clearly demonstrates the effectiveness and power of cross-modal learning in distinguishing real and synthetic images. Table II shows the hyperparameter values for our fusion model. As illustrated in Figure 2, the t-SNE visualization further confirms that the fused features form clearly separable clusters for AI-generated and real samples, highlighting the model's ability to learn discriminative multimodal representations.

The fusion framework combines semantic cues from text and pixel-level visual features, which increases contextual perception and interpretation and reduces the limitations of the unimodal architecture. Although the multimodal model shows a slight decrease in recall compared to the text-baseline, the addition of visual information makes the model more conservative; that is, evidence from both methods is required to flag an instance. Irrespective of this trade-off, the model has a high accuracy of (96%). This is important when it comes to real-news verification scenarios because the false alarms of real news that were incorrectly marked because of unusual writing styles are reduced. Cross-attention layers are useful in matching visual components with textual assertions, thereby effectively in filtering out unimodal false positives. Nonetheless, this is accompanied by a certain performance loss in the isolated test sets that might be attributed to differences in domains, such as texture, light, or style of rendering. Such discrepancies point to the importance of having the correct integration between text and image embeddings when it comes to creating the best generalization. In general, the multimodal fusion approach has a more detailed and more accurate representation of the image, which is better at retraction and contextual sensitivity than when it comes to single-modality systems, and provides a solid foundation on which future developments in cross-modal synthetic image detection will be built.

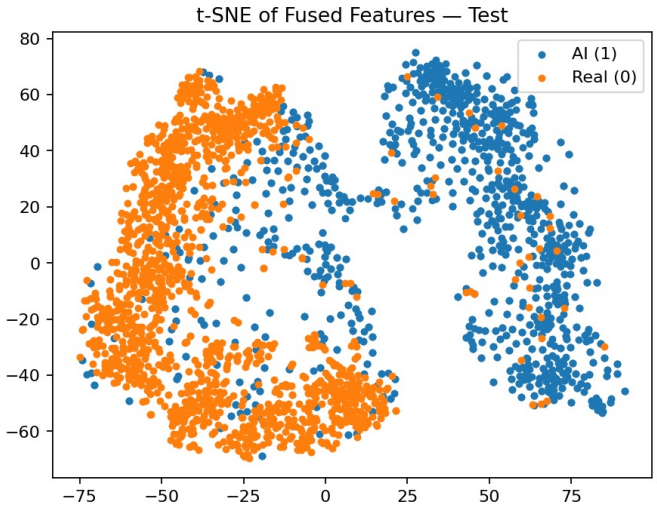

Fig. 2. t-SNE visualization of fused multimodal features on the test set. The plot shows clear separation between AI-generated samples (blue) and real samples (orange), demonstrating the model's discriminative ability.

D. Comparative Analysis

To ensure a fair comparison, Table I includes only baselines with similar fusion strategies and backbone scales for news-style corpora. We exclude alignment-pretrained CLIP variants

TABLE I
COMPARISON OF MULTIMODAL FAKE-NEWS DETECTION METHODS

Dataset	Model	Modalities Used	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Weibo, Twitter [25]	CLIP + Fine-tuned Transformer	Text + Image	93.5	92.1	93.0	92.5
FakeNewsNet [26]	CNN (Image) + BERT (Text)	Text + Image	90.2	89.5	88.7	89.1
Twitter, Weibo [27]	BERT + VGG-19 with Attention Fusion	Text + Image	∼88	87.1	86.5	86.8
PolitiFact, GossipCop [28]	CNN–RNN Multimodal Fusion	Text + Image	91.2	90.5	89.8	90.1
FakeNewsNet [29]	BERT + ResNet + Semantic Graph Fusion	Text + Image	92.4	91.7	91.0	91.3
Weibo [30]	Multimodal BERT with Pyramid Attention	Text + Image	94.1	93.4	92.9	93.1
MiRAGeNews	**Cross-modal Fusion (ours)**	**Text + Image**	**0.89**	**0.96**	**0.80**	**0.88**
MiRAGeNews	**RoBERTa (Text only)**	**Text**	**0.85**	**0.78**	**0.97**	**0.87**
MiRAGeNews	**ViT (Image only)**	**Image**	**0.73**	**0.68**	**0.89**	**0.77**

TABLE II
HYPERPARAMETERS OF THE FUSION MODEL

Parameter	Value
Encoders	RoBERTa-base + ViT-B/16
Trainable/Frozen layers	Bottom 8 frozen, top 4 fine-tuned
Hidden size	768
Cross-modal attention	2 blocks, 8 heads
Fusion	Concat (CLS' + TEXT_SUM'), MLP 1536→512→256→128→1
Loss	BCEWithLogits + label smoothing (0.1)
Optimizer	AdamW (weight decay 0.001)
Learning rates	5e–6 (encoders), 5e–5 (fusion)
Scheduler	Cosine LR with 20% warmup
Batch size	16

and deep co-attention stacks trained on different distributions. Among these similar sets - multimodal CNN+BERT, BERT+VGG-19 with attention, multi-level fusion CNN-RNN, and semantic-enhanced MFN - our fusion model achieved 0.8856 accuracy and 0.8762 F1-score. Notably, it delivers much higher precision at 0.9547 and stronger overall discrimination with an ROC-AUC of 0.9411. The gain in precision is substantial, exceeding these baselines by three to eight percentage points, while F1 remains competitive despite MiRAGeNews being the most challenging dataset in our study. These results show that late fusion, combined with conservative regularization and moderate augmentation, can yield high-certainty decisions without relying on heavy alignment modules or dataset-specific attention stacks.

When compared with our unimodal references on the same test split, the fusion system improves accuracy by 0.0388 over the text-only model and by 0.1520 over the image-only model. This increases the F1 score by 0.1084 compared to the image-only model, while maintaining a high level of accuracy compared to the curated baseline. This implies that the model has a relatively lower number of false alarms and offers more stable calibration by distribution, which is a crucial property to be used in news verification pipelines. Comprehensively, our model is stable as a strong multimodal detector in the benchmark set despite its limitations in the comparative baseline and settings that are sensitive to accuracy.

V. DISCUSSION

According to the analysis of CIVA-Net, multimodal fusion is an effective means of adding performance to fake news detection, particularly when it comes to AI-generated high-quality content. According to experimental evidence, CIVA-Net can maintain high accuracy in the various evaluation measures and show a stable performance. The notable point is that it delivers much better results than unimodal baselines, which proves that the combinatory use of text and visual cues also finds significant information that would have otherwise remained unnoticed.

These results are consistent with the results of previous multimodal detection experiments that demonstrated better performance when language and visual data are used together. Nonetheless, CIVA-Net stands out because it has a sophisticated cross-modal attention system that identifies minor semantic-visual associations between captions and images. This enables even subtle AI-generated forgery to be picked up by the model. To illustrate, the unimodal baselines did not work, but CIVA-Net was able to classify the cases that had seemingly harmless captions and clear manipulated images correctly. This is the reason this model has a clear feature-space clustering and has a high overall performance.

A key aspect of our development process was to successfully prevent overfitting, a common challenge in multimodal learning. We applied regularization and dropout techniques, so that the excellent performance of the model is not only due to remembering the training data, but also demonstrates proper generalization to the test dataset. The observed performance drop on the unseen test sets suggests that the cross-attention mechanism is sensitive to the specific rendering artifacts of different generators. While the attention layers learn to spot semantic mismatches, they may also overfit to the specific 'glossy' texture of Midjourney or DALL-E. Future work must address this by applying stronger data augmentation during the fusion stage to force the model to focus less on pixel-level artifacts and more on the structural alignment between text and image.

However, challenges remain, particularly concerning generalization. The relatively lower recall, while improved over baselines, indicates that certain highly nuanced or borderline cases of misinformation continue to evade detection. We hy-

pothesize that this is a limitation tied to the inherent ambiguity of some human-created fakes or the extreme subtlety of novel AI-generated artifacts not represented in the training distribution. Moreover, the observed weaker performance of the ViT component on out-of-domain samples points toward a sensitivity to dataset bias. This shows that this model is vulnerable to variations in image type, lighting, or content that are not in the training domain.

These hints have significant theoretical and practical implications. Conceptually, this confirms the contribution of the cross-modal attention as an effective mechanism of identifying the presence of semantic-visual discrepancies and forms a basis for future multimodal detection systems. In practical terms, these findings indicate good system design in the management of the ever-increasing complex information manipulation in the real world. The future studies must focus on utilizing adaptive training techniques to enhance the domain robustness using adversarial-based training techniques and making models learn as they mature to new forms of misinformation. Moreover, it is also necessary to investigate lightweight and efficient architectures in the dynamically evolving news verification setting.

VI. Conclusion

This paper introduces a new multimodal model, CIVA Net, a cross-modal attention-based AI-generated misleading information detection framework that refers to the analysis of both text and visual characteristics in a coordinated fashion. The new and challenging MiRAGeNews dataset that was used in the study underwent experiments that revealed that the proposed model has a consistent performance as opposed to that of unimodal and existing multimodal methods. These findings all point to the fact that the cross-modal attention mechanism of CIVA-Net is able to effectively identify minor semantic discrepancies between text and visual items, which is a key indicator to recognize AI-generated misleading information. The primary value of the research paper is that it offers a stable and evidence-based model of multimodal fake news detection, which can be successfully implemented in practical social media and digital news systems. Although the experimental the promising results are promising, one can still achieve the purpose of also improving the recall efficiency and generalization in the model across domains as one of the things that can guide research in the future.

Generally, this research work also proves the efficiency of. The multimodal fusion methods suggest that further research must not be confined to certain dataset-based performance, but it must be aimed at creating a domainless and real-time adaptive model that would be able to adjust to the dynamically evolving generative media landscape. This is a necessity in the preservation of the integrity of digital information in the generative AI era.

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